Documention of Lego

- What was your experience?
  - Did you find this an easy way to document your process?
  - Was there anything you found difficult to grasp?
  - Did those who tried to replicate it find it easy to follow the assembly instructions?
  - Have you come across anything in the instructions that is unclear?

Documention of Lego

- Conclusion:
  - There are multiple ways to document
  - Some common structures (Template, List of Bricks, Fotos) may help for documentation
- So now – let’s try to structure the possibilites of data documentation!

Levels of data documentation



1 **Study-level documentation**

overall **context** of the research

2 **Data-level documentation**

describes **individual** data **files**

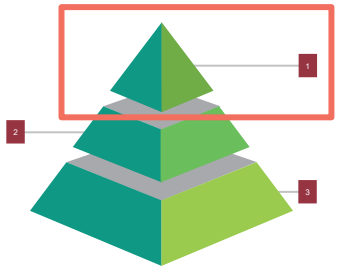
3 **Metadata**

standardised, structured data about the data

Source: Wolff, Musloff 2024 (FDM-BB Zertifikatskurs) modified by FAIRagro 2024

Study-level Documentation

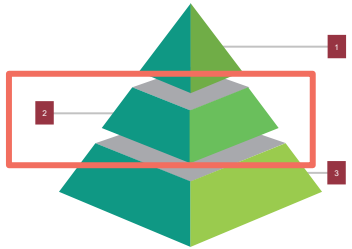
- **WHY** was the data generated?
  - → Research project description, goals, hypotheses...
- **WHAT** is the content of the data?
  - → Structure & relationship of the files to each other
- **HOW, WHERE** and **WHEN** was the data generated? & **WHO** generated the data?
  - → Data collection information
- **HOW** was the data processed?
  - → Describe data processing
  - → Versioning
- Information on access and terms of use.



- Tools to use:
  - → **DMP**
  - → Lab/Field Notebooks
  - → **README file**
  - → SOPs
  - → Lab Protocols
  - → User Guides

Source: UK Data Service and Wolff, Musloff (FDM-BB Zertifikatskurs) 2024 modified by FAIRagro

Data-level Documentation



- **Information** on the data files:
  - Data & file type, format, size, data processing scripts ...
- **Explanation** of variables, labels and codes:
  - Names, labels, values ...
- Tools to use:
  - **README file**
  - Data dictionary
  - Codebooks
  - Commented Code
  - Lab/ Field Notebooks

```
GENERAL INFORMATION

G01. Names of file(s) or dataset(s) that this README file describes
G02. Date of creation/last update of the README file
G03. Name and contact information of Principal Investigator
G04. ORCID of Principal Investigator
G05. Institution of Principal Investigator
G06. Contact of other person at KU Leuven that has access to the dataset
G07. Description of the dataset
G08. Keywords (author defined)
G09. Thesaurus or controlled vocabulary keywords
G10. Thesaurus or controlled vocabulary used in this README
G11. Language(s) used in the dataset
G12. Other involved researchers

PROJECT INFORMATION

P01. Project information
P02. Project abstract
P03. Project funder: Name of funder, type of grant, grant number

FILE OVERVIEW

F01. Number of files described by the README-file
F02. List with names of files, description, date of creation of file
F03. File formats
F04. Software used to generate the data
F05. Software necessary to open the file
F06. Relationship between the files
F07. Which version of the dataset is this? Date of this version?
F08. Information about the dataset versions and reason for updates
F09. Naming conventions for file names

STORAGE INFORMATION

S01. Where are the data stored?
S02. Links to other available locations of the dataset (e.g. repository)
```

Source: UK Data Service and CESSDA Training Team (2017 - 2022) modified by FAIRagro 2024,; Illustration: Screenshot template from KU Leuven 2024 (public domain)

Metadata

"Data about data" -> differentiation from primary data!

- **Standardised** & **structured** information
- **Human-** and **machine-readable**
- Goals:
  - describes, explains, locates
  - makes it easier to retrieve, use & manage



Source: Wolff, Musloff 2024 (FDM-BB Zertifikatskurs) modified by FAIRagro 2024

Importance of Metadata

- Makes data:
  - **Findable**, **Accessible**, **Interoperable** and **Reusable** (**FAIR**)
  - **Readable** and **understandable**
  - **Citable**
  - More easily **shareable** across an organisation
  - **Consistent**
- Facilitates the **long-term archiving** of data
- Makes your data more **reproducible**



Source: Lomax 2019 modified by ZB MED 2022 and FAIRagro,; Illustration: Image by The Turing Way Community & Scriberia (2023) is licensed under CC BY 4.0

Types of metadata

Bibliographic	Administrative	Process	Content-describing/descriptive
Identifier of the resource, **title**, **author**, date of publication, subject, publisher, ...	File types, file names, **formats**, location (file path), **unique ID**, access rights, licence, ...	Used methods, used sensors, **wavelength**, depth of measurement, ...	Crop, genotype information, **variety**, breeding line or variation, environmental conditions of the study, type of climate, **type of soil**, time of year, Geolocation, Treatment - Abiotic (e.g. light, temperature, soil, ...) - Biotic (e.g. animals, fungi, ...) - Crop management (e.g. application of chemicals, crop rotation, density, ...), (Observed) Traits, ...

Source: Engelhardt et al. 2022 and forschungsdaten.info modified by FAIRagro 2024

FAIRagro

Metadata Standards

Definition of Metadata Standards

Collections of metadata elements that are used for a **specific purpose** (e.g. description of digital objects, description of books, ...)

They are also “formal and machine-readable structure that has been established by the large research community or by an international standard organization” [KU Leuven 2022].



Created by Dewi Novita Sari
from Noun Project

Source: KU Leuven 2022 modified by ZB MED 2024; Illustration: “definition” by Dewi Novita Sari is licensed under CC BY 3.0

Importance of Metadata Standards

- Agreement on a standardised definition of terms leading to a **common understanding**
  - Promoting **integration** in the community that follow the standard you choose
- Making data **F**indable, **A**ccessible, **I**nteroperable and **R**eusable (**FAIR**)

Source: Modified from KU Leuven 2022 by ZB MED 2024

Benefits and Drawbacks of Metadata Standards



- Agreement on a standardised definition of terms leading to a **common understanding**
- Data can be more easily **combined** and **exchanged**
- Guidance on what information is **important** in documenting your data



- **Time** and **know-how** needed to understand and apply a metadata standard
- Less flexibility in documenting your data as you have to follow the principles of a standard

Source: Modified from KU Leuven 2022 by ZB MED 2024

Metadata standards & documentation

ISO 19115 - international standard for **geographic** information

INSPIRE - european standard for **geographic** information

Darwin Core - standard for information about **biological diversity**

MIAPPE - Minimum Information About **Plant Phenotyping** Experiments

Source: RDA Metadata Standards Catalog

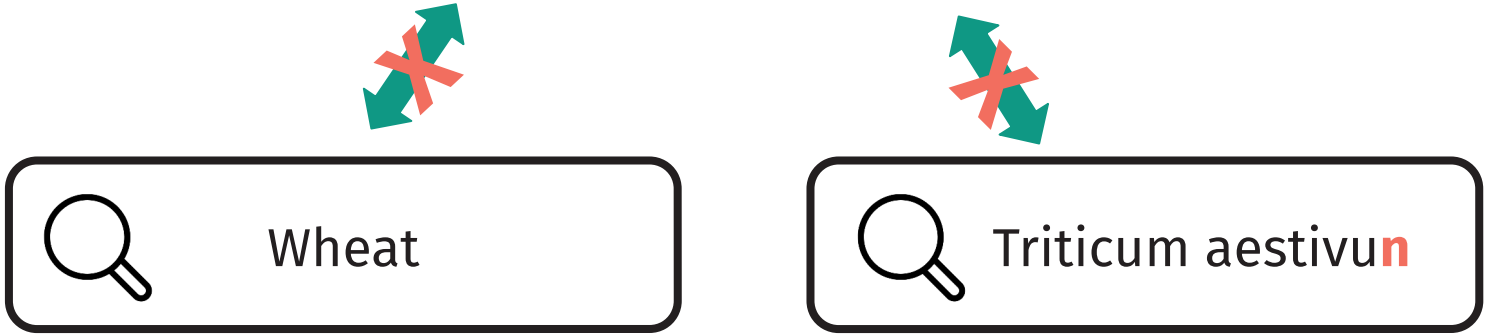
FAIRagro

Taking it one step further:

How to increase metadata interoperability

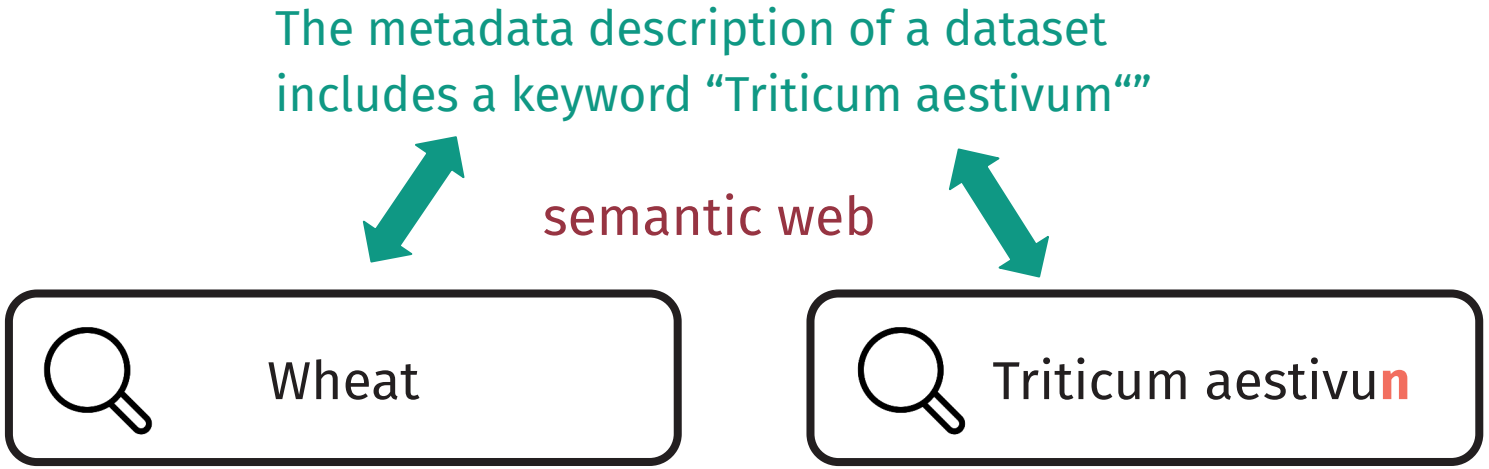
Machine understandability of metadata

The metadata description of a dataset includes a keyword “Triticum aestivum“”



Source: Gabriel Schneider 2024 modified by FAIRagro in 2024

Machine understandability of metadata



Source: Gabriel Schneider 2024 modified by FAIRagro in 2024

How to document?

Standards for suitable **documentation:**

- Controlled vocabularies such as AGROVOC
- Soil classification systems -> Bodensystematik, FAO's World Reference Base, USDA's Soil Classification
- Gene classification -> Gene Ontology
- Agricultural Data -> Crop Ontology

Illustrations: Logos of Gene Ontology, Crop Ontology are licensed under CC BY 4.0

AGROVOC

- Multilingual semantic resource collecting concepts, terms, definitions and relationships between them
- Coordinated by the Food and Agriculture Organization of the United Nations (FAO) since the 1980s
- International editor cooperation
- +41 200 concepts
- +1 177 000 terms
- up to 42 languages

Source: AGROVOC

AGROVOC

- Relations to other concepts and terms in AGROVOC → enables searches like “show me all pathogens related to Triticum aestivum”

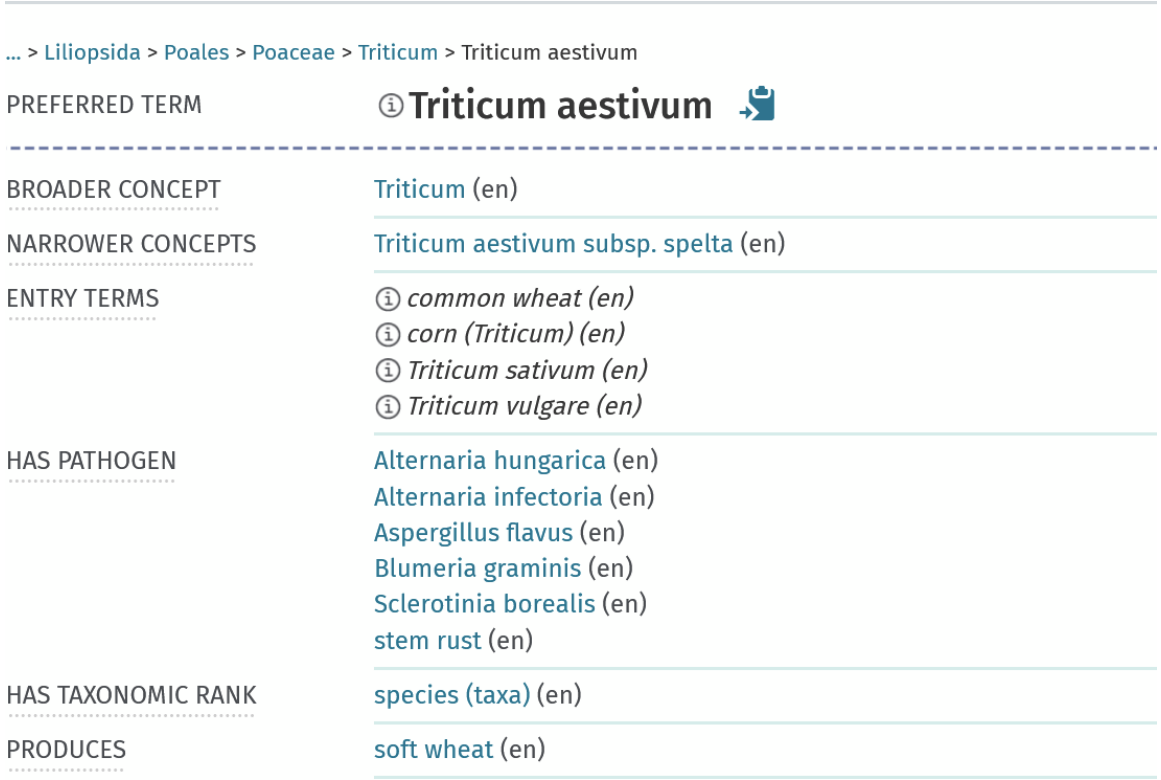


Illustration: Screenshot AGROVOC from 23.04.2026, public domain

AGROVOC

- Names of the same concept in different languages → enables multilingual searches



Illustration: Screenshot AGROVOC from 23.04.2026, public domain

AGROVOC

- Persistent identifier of the term

URI	http://aims.fao.org/aos/agrovoc/c_7951

- Mappings to other semantic resources

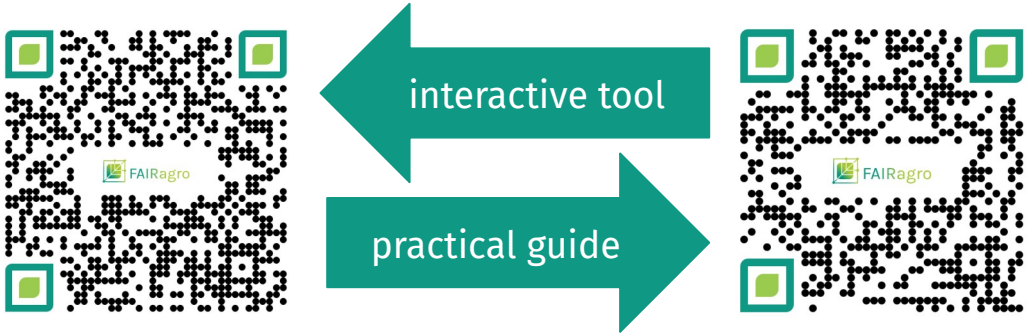
CLOSELY MATCHING CONCEPTS	http://dbpedia.org/resource/Triticum_aestivum	dbpedia.org
EXACTLY MATCHING CONCEPTS	https://lod.nal.usda.gov/nalt/66457	lod.nal.usda.gov

Illustration: Screenshot AGROVOC from 23.04.2026, public domain

How to generate Metadata in Practice

The metadata decision tree guides you through the **key questions for data documentation**.

It helps you **identify suitable data management practices** according to your specific needs.



What will be your next steps in documenting your data?